```
model MinCM2.mod; data net.dat; options solver minos;
param alfa;  param u{links};  param vv{links};

param pSTOP;  param l;  param relgap;
param f_obj;

problem AoN: v_k, v, N, flux_total, Vg;
problem Q:   v_k, v, N, flux_total, Vnl;

let {(i,j) in links} t0[i,j]:=1;
solve AoN;
let {(i,j) in links} vv[i,j]:=v[i,j];

let pSTOP:=0;
let l:=1;
repeat while l <=100 and pSTOP=0 {
    let {(i,j) in links} t0[i,j]:=c[i,j] + cap[i,j]*vv[i,j];
    solve AoN;
    let relgap:= -(sum{(i,j) in links}(t0[i,j]*(v[i,j]-
vv[i,j])))/(sum{(i,j) in links}t0[i,j]*vv[i,j]);
    if relgap <= 0.01 then let pSTOP:=1;
    let alfa:= -(sum{(i,j) in links}(t0[i,j]*(v[i,j]-
vv[i,j])))/(sum{(i,j) in links}cap[i,j]*(v[i,j]-vv[i,j])^2);
    let {(i,j) in links} vv[i,j]:=vv[i,j]+alfa*(v[i,j] - vv[i,j]);
    let f_obj:= sum{(i,j) in links} (c[i, j]*vv[i, j] +
0.5*cap[i,j]*vv[i,j]^2);
    let l:=l+1;
}
```

```
display vv;
solve Q;
display {(i,j) in links}: v[i,j], vv[i,j]-v[i,j];
```